

SSV Mission Simulator - can this vessel survive a voyage from [origin] to [destination]? If it can, find the cheapest material configuration needed to complete the mission.

Building autonomous ocean vessels is a billion-dollar industry. Companies like Saronic (now valued at \$9.25 billion after recently awarded a \$392M Navy contract for its Corsair autonomous surface vessel) are changing Naval warfare. But current testing validates whether a vessel can withstand conditions - not whether it can actually complete a mission. Running out of fuel, losing structural integrity, or capsizing never shows up in a conditions test.

We built a simulation engine that lets anyone answer the question that matters before a single vessel hits the water: Can this hull configuration survive a real mission voyage from point A to point B - and what's the cheapest material configuration needed to make sure it does.

We run thousands of simulations - each one a voyage attempt with a different material configuration, hull geometry, and environmental profile. The AI learns from every failure - which weld cracked first, which seal gave out, at what sea state the hull buckled - and iterates toward the cheapest configuration that survives.

#### Layers

1. Simulation engine
  - a. physics, wind, waves, ice accretion, salinity & pH (maybe moving obstacles later on)
2. Ship (ship properties, ship state, how the ship interacts with the engine)
3. Model (simulates the ship in the simulation engine thousands of times until it can get successfully complete the mission from point a to point b with the cheapest materials possible and shortest time through self-learning starting from scratch (AI has no prior knowledge of how anything works))

#### Example usage

1. Saronic engineer drops the vessel's spec sheet into the simulation engine
2. Choose the origin and destination locations
3. Agent loop (failure diagnosis → targeted parameter revision → re-simulate)
  - a. Run simulation
  - b. Evaluate why it failed, which parts broke first, etc
  - c. Revise parameters based on failure diagnosis
  - d. (loop until max iterations / convergence threshold is met)
4. Once a surviving configuration is found, return the cheapest one

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## Simulator

- Hull geometry, material grade, weld quality, seal state, crack state
- Waves (height, period, slamming probability)
- Water temp + salinity (corrosion -> fatigue -> crack)
- GM state (capsize check)
- Fuel state
- Failure endings: sink, capsize, breaks in half

## Eval state

- Distance completed (% of voyage)
- Time to first failure
- Which component failed first
- Severity of each failure mode (crack length, GM margin, fuel remaining)
- Total material cost of current configuration